



Mamba-Powered Pavement Safety

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Introduction

Highways play a vital role in transportation, economic development, and public safety. However, over time, asphalt pavements are prone to developing potholes due to continuous traffic load, weathering effects, and material deterioration. These potholes not only reduce road quality and driving comfort, but also increase the risk of accidents and can cause vehicle damage such as tire punctures, dents, and leaks. Therefore, regular and accurate detection of potholes is essential for effective highway maintenance and for ensuring safe road operation.



Problem Statement

Current road inspection methods largely depend on manual surveys, which are time-consuming, labor-intensive, and prone to human error, making it difficult to ensure consistent and timely detection of pavement damages such as potholes. Although machine learning and deep learning techniques have improved automated detection, challenges such as image distortion from oblique camera angles, limited feature extraction, and reduced detection efficiency still affect the accuracy and reliability of these systems. This project addresses the need for an efficient and accurate automated pothole detection framework that utilizes advanced image correction and deep learning-based segmentation techniques to improve detection performance and support safer and more effective road maintenance.



Objectives

- To develop a robust deep learning pipeline for accurately segmenting road damage from input road images.
- To integrate perspective correction to handle image distortion and improve geometric consistency of detections.
- To introduce a novel ShadowNet module for suppressing shadow effects, improving performance under varying lighting conditions.
- To build a modular system supporting training, inference, and comparative evaluation of model variants.



Expected Outcomes

- Accurate segmentation results with improved IoU and Dice scores, producing precise damage masks.
- Reduced geometric distortion errors, resulting in more reliable area estimation and spatial localization.
- Improved robustness to shadows and lighting variations through the integration of ShadowNet, reducing false detections.
- A complete and reproducible workflow with efficient inference, model evaluation, and multi-variant comparison capabilities.



Literature Survey Number -1

Zhang et al. (2023) proposed a deep learning–based method for automatic pothole detection using image segmentation techniques. The study focused on improving road damage identification by applying convolutional neural networks to extract pavement features from road images.

Merits:

- Provides high accuracy in detecting potholes from complex road images.
- Reduces the need for manual road inspection and improves monitoring efficiency.

Demerits:

- Performance can be affected by poor lighting conditions and shadows.
- Requires a large dataset of labeled road images for effective training.

Reference: Zhang et al., “Pavement Pothole Detection Using Deep Learning-Based Image Segmentation,” 2023.



Literature Survey Number -2

Li et al. (2022) conducted research on pavement damage detection using deep learning models combined with image preprocessing techniques.

Merits:

- Improves detection accuracy by correcting image distortion before analysis.
- Enhances feature extraction for better identification of pavement damage.

Demerits:

- Image preprocessing increases computational complexity.
- Detection performance may vary depending on camera position and image quality.

Reference: Li et al., “Automatic Road Damage Detection Using Image Correction and Deep Learning Segmentation,” 2022.



Literature Survey Number-3

Maeda et al. (2018) conducted research on road damage detection using deep learning and image-based datasets. They developed a system that uses convolutional neural networks (CNNs) to automatically identify different types of road damages.

Merits:

- Enables automatic and large-scale road damage detection using smartphone or vehicle cameras.
- Provides a publicly available dataset that helps improve research in road damage detection.

Demerits:

- Detection accuracy may decrease in complex environments with shadows or poor lighting.
- Requires high computational resources for training deep learning models.

Reference: K. Maeda et al., “Road Damage Detection Using Deep Neural Networks with Images Captured Through a Smartphone,” *Computer-Aided Civil and Infrastructure Engineering*, 2018.



Literature Survey Number -4

Koch and Brilakis (2011) developed an automated pavement defect detection system using computer vision techniques. Their approach used image processing algorithms to identify potholes and cracks from road surface images captured by cameras.

Merits:

- Reduces the need for manual road inspection.
- Can detect multiple types of pavement defects.

Demerits:

- Performance depends heavily on image quality.
- Traditional image processing methods have limited accuracy compared to deep learning.

Reference: C. Koch and I. Brilakis, “Pothole Detection in Asphalt Pavement Images,” *Informatics*, 2011.



Literature Survey Number -5

Zhang et al. (2019) proposed a deep learning-based pothole detection method using Convolutional Neural Networks (CNNs) to automatically classify road surface images and detect pothole regions.

Merits:

- High accuracy in detecting potholes in complex road environments.
- Capable of learning important road surface features automatically.

Demerits:

- Requires large labeled datasets for training.
- Computationally expensive for real-time applications.

Reference: L. Zhang et al., “Road Damage Detection Using Deep Convolutional Neural Networks,” 2019.



Literature Survey Number -6

Fan et al. (2020) introduced a road damage detection system using deep learning and object detection models such as YOLO to detect potholes and cracks in real-time.

Merits:

- Provides fast detection suitable for real-time monitoring.
- Capable of detecting multiple road defects simultaneously.

Demerits:

- Accuracy may reduce under poor lighting conditions.
- Requires high-performance hardware for deployment.

Reference: R. Fan et al., “Road Damage Detection Based on Deep Neural Networks,” *IEEE Intelligent Transportation Systems*, 2020.



Literature Survey Number -7

Gopalakrishnan et al. (2017) proposed a deep learning-based pavement distress detection system using CNN models trained on road surface images to identify potholes and cracks.

Merits:

- Achieves high detection accuracy using deep learning.
- Improves efficiency of pavement condition monitoring.

Demerits:

- Training requires large datasets and computational power.
- Performance may vary with environmental conditions.

Reference: K. Gopalakrishnan et al., “Deep Learning-Based Pavement Distress Detection,” *Transportation Research Board*, 2017.



Literature Survey Number -8

Maeda et al. (2018) developed a road damage detection system using smartphone images and deep neural networks to detect different types of road damages including potholes.

Merits:

- Enables large-scale road monitoring using smartphones.
- Provides a publicly available dataset for research.

Demerits:

- Detection accuracy may decrease in shadowed or unclear images.
- Requires stable image capture for reliable detection.

Reference: K. Maeda et al., “Road Damage Detection Using Deep Neural Networks with Smartphone Images,” *Computer-Aided Civil and Infrastructure Engineering*, 2018.



Literature Survey Number -9

Arya et al. (2020) proposed a pothole detection system using image processing and machine learning techniques to identify potholes from road surface images.

Merits:

- Simple and cost-effective detection approach.
- Reduces manual inspection workload.

Demerits:

- Limited accuracy compared to deep learning models.
- Sensitive to variations in lighting and road texture.

Reference: D. Arya et al., “Deep Learning-Based Road Damage Detection and Classification,” 2020.



Literature Survey Number -10

Silva and Jung (2021) introduced a road surface monitoring system using deep learning segmentation models to precisely locate pothole regions in pavement images.

Merits:

- Provides accurate segmentation of pothole areas.
- Helps in detailed pavement damage analysis.

Demerits:

- Segmentation models require high computational resources.
- Training complexity is higher compared to simple detection models.

Reference: S. Silva and C. Jung, “Real-Time Road Surface Monitoring Using Deep Learning Segmentation,” 2021.



Literature Survey

Consolidation of the literatures:

S. No.	Author(s) & Year	Methodology used	Limitations
1	Gadekar et al. (2024)	Sonar-assisted YOLO Model.	Sonar integration adds sensor complexity and calibration needs.
2	Lakshminarayanan et al. (2024)	CNN + IoT-based Cloud Reporting	Latency issues in regions with poor network connectivity.
3	Khan et al. (2024)	YOLOv8-based Pipeline	Struggles with "false potholes" caused by complex road shadows.
4	Zanevych et al. (2024)	YOLOv7 & YOLOv8 (Low-light)	Still faces significant accuracy drops in complete darkness.



S.No.	Author(s) & Year	Methodology used	Limitations
5	Reddy et al. (2024)	Custom Deep Learning Model.	Model optimization for mobile edge devices remains a challenge.
6	Vinodhini et al. (2024)	Transfer Learning + CNN	Requires a massive and diverse dataset for generalizability.
7	Safyari et al. (2024)	Comprehensive YOLO Review	Existing models still struggle with varying road textures.
8	Yao et al. (2025)	Binocular Vision + Deep Learning	High hardware cost compared to monocular camera systems.



S.N o.	Author(s) & Year	Methodology used	Limitations
9	Boosam et al. (2025)	Comparative study: YOLOv1 to YOLOv5 vs. SSD	High dependence on high-quality annotated training data.
10	Chandra et al. (2025)	YOLOv5 on Drivable Area Dataset	Performance drops under low-light or night conditions.

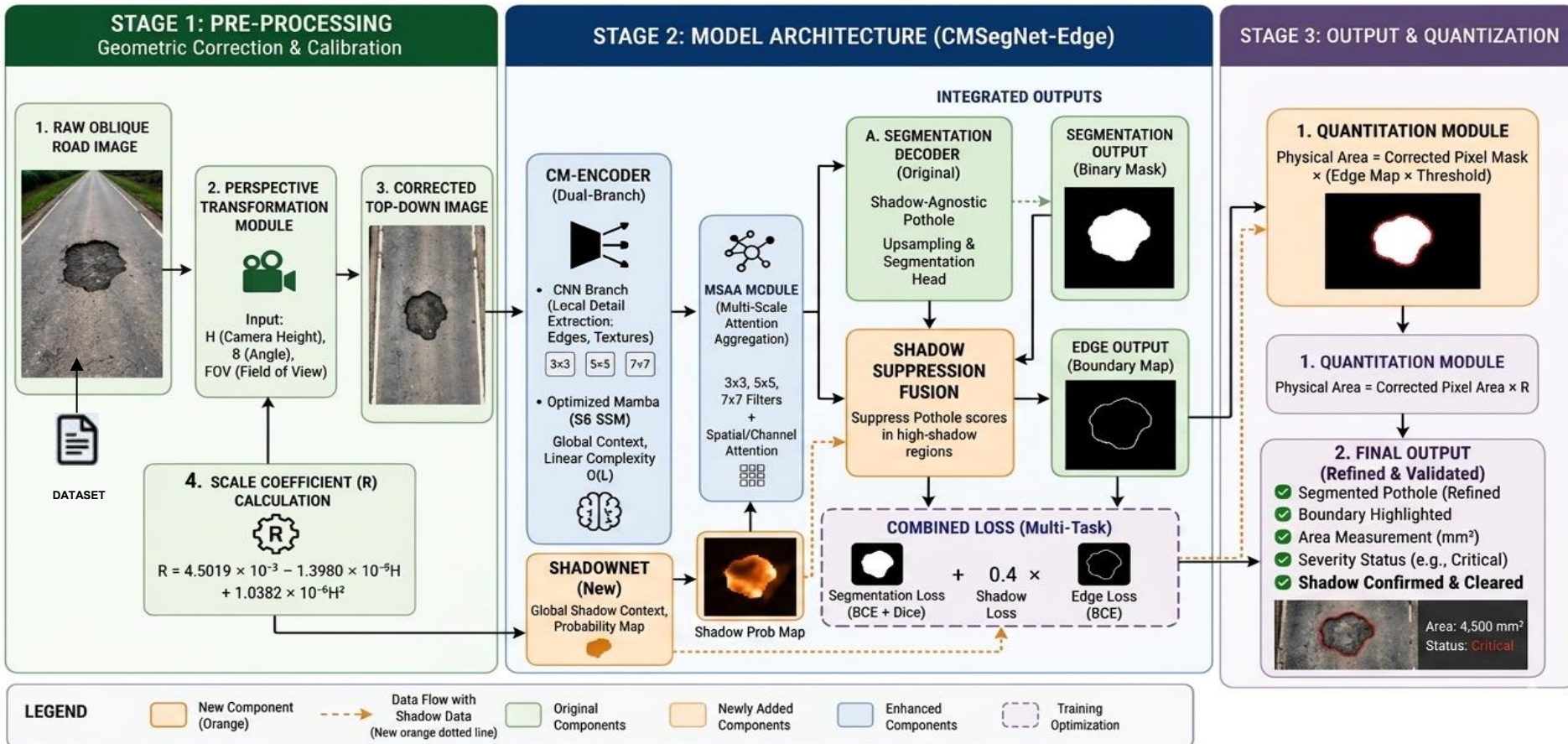


Dataset Details

S.No.	Dataset Name	Description	Purpose
1	Mendeley Road Distress (Passos, 2020).	2,235 Images : High-resolution images from Brazilian highways with multiple defect masks.	The datasets provide Pixel-level Binary Masks , which are essential for training the Mamba-based segmentation engine.

System Design Architecture

CMSegNet-Edge: Enhanced with Shadow Suppression





Model Description

1. CM-Encoder (Convolution-Mamba Encoder)

The CM-Encoder integrates convolutional modules and optimized Mamba modules at each encoder stage. The convolutional branch extracts high-resolution local spatial features of pavement damage using three residual convolution blocks: 1×1 channel compression, 3×3 spatial convolution, and 3×3 dilated spatial convolution. The optimized Mamba module, built on a Structured State Space Model (SSM), captures long-range global context with linear time complexity $O(L)$, avoiding the $O(L^2)$ bottleneck of Transformer self-attention. Channel and spatial attention sub-modules enhance feature selectivity, and the two branches are merged through element-wise multiplication and projected back to the original channel dimension.



2. Multi-Scale Attention Aggregation (MSAA) Module

The MSAA module enriches encoder output feature maps by fusing multi-scale local and global information. Parallel 3×3 , 5×5 , and 7×7 convolutions perform multi-scale feature extraction, followed by spatial aggregation through 7×7 convolution with sigmoid gating, and channel aggregation using global average pooling with two 1×1 convolutional layers. The fused output enhances both spatially and channel-wise important features, improving discrimination between true potholes and visually similar regions such as debris or aggregate patches.

3. Lightweight Feature Decoder

A lightweight MLP-based decoder is employed to reduce computational cost. Multi-scale features from the MSAA module are combined through 1×1 convolution, upsampled, and processed by 8×8 convolution. The merged features are mapped to 1-D tokens, and spatial and channel modeling is performed via MLP layers. A final upsampling step produces the pixel-level segmentation mask predicting pothole regions at the original image resolution.



4. ShadowNet - Shadow Suppression Module

ShadowNet is a dedicated shadow detection sub-network integrated into the full inference pipeline. Road shadows from trees, buildings, and vehicles frequently create dark regions that resemble potholes in color and texture, leading to false positives in conventional models.

ShadowNet generates a probability map of shadow regions from the input image. This map is fused with CMSegNet-Edge's segmentation output to suppress false detections in shadowed areas. A safety fallback mechanism ensures that if ShadowNet's output appears near-constant (indicating unstable prediction), inference reverts to the CMSegNet-Edge mask alone.

Visual Analysis



CMSegNet

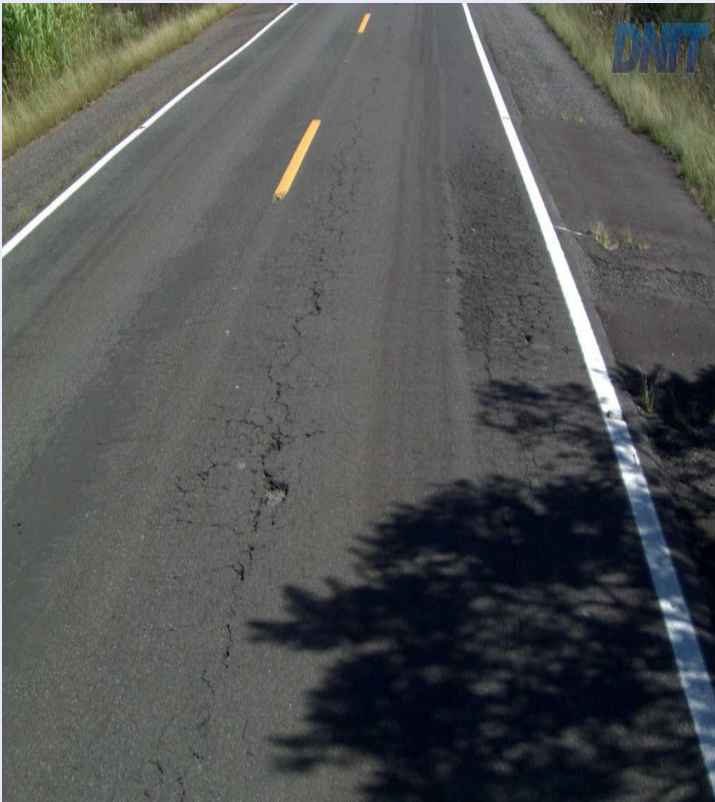


CMSegNet + EdgeHead



CMSegNet + ShadowNet

Implementation Result



Original Photo



Pothole Identified

Evolutional Metrics

Model Version Comparison - Detailed Metrics Summary

Metric	CMSegNet	CMSegNet + EH	CMSegNet + ShadowNet
Accuracy	0.973454	0.978177	0.963228
Precision	0.124055	0.149333	0.131066
Recall	0.166809	0.139090	0.317187
Specificity	0.984244	0.989401	0.971870
F1	0.142290	0.144030	0.185486
IoU	0.076594	0.077604	0.102223
Dice	0.142290	0.144030	0.185486
MCC	0.130566	0.133078	0.187600



Purposed Novelty

- Adds explicit class-imbalance handling to lift recall/IoU beyond the CMSegNet training setup.
- Introduces ShadowNet-based shadow suppression with a reliability fallback to reduce shadow-driven false positives during inference.
- Provides a unified multi-model comparison pipeline that outputs CSV metrics plus 6 visualizations for fast model selection.
- Delivers a reproducible end-to-end workflow .



Working Environment

Hardware:

- RTX 3050 GPU
- 16GB RAM
- Windows 11

Software:

- Python
- TensorFlow / PyTorch
- Open CV
- VS Code



Social Impact

Improvement in Road Safety:

- The proposed pothole detection system has a significant positive impact on road safety by enabling early identification and timely repair of road surface damages. Potholes are one of the major causes of road accidents, especially for two-wheelers and small vehicles, as they can lead to sudden loss of control, vehicle damage, or even fatal injuries.

Efficient Road Maintenance and Cost Reduction:

- Another important social benefit of the system is its ability to improve the efficiency of road maintenance operations. Traditional methods of road inspection are manual, timeconsuming, and often inefficient, leading to delayed repairs and increased maintenance costs.

Support for Smart City Development:

- The integration of the pothole detection system with modern technologies such as IoT and cloud computing supports the development of smart cities. By enabling real-time data collection and analysis.



Future Enhancement

Real-Time Implementation and Edge Deployment:

- One of the major future enhancements of the proposed pothole detection system is its deployment in real-time environments using edge devices such as mobile phones, embedded systems, or vehicle-mounted cameras.

Dataset Expansion and Generalization:

- Another important enhancement involves expanding the dataset used for training the model. The current system may be limited by the diversity of training data, which can affect its ability to generalize to new environments.

Multi-Damage Detection System:

- The current system is primarily focused on detecting potholes; however, road surfaces often suffer from multiple types of damage such as cracks, patches, and surface wear. A future enhancement would be to extend the system to detect and classify multiple categories of road damage within a single framework.



Future Enhancement

Integration with IoT and Smart City Systems:

- Integrating the proposed system with Internet of Things (IoT) technology and smart city infrastructure can significantly enhance its practical usability. In this approach, detected pothole data can be transmitted in real time to a centralized cloud platform using connected devices.

Advanced Analytics and Decision Support System

- Another promising enhancement is the development of an advanced analytics and decision support system based on the output of the pothole detection model. Instead of simply identifying potholes, the system can be extended to analyze patterns such as frequency of occurrence, severity levels, and high-risk zones.



Conclusion

- The proposed pothole detection and segmentation system presents a robust and efficient solution for identifying road surface damages using advanced deep learning techniques. By integrating a multi-stage pipeline consisting of preprocessing, CMSegNetbased segmentation, auxiliary prediction, and shadow suppression through Shadow SegNet, the system effectively addresses key challenges present in real-world environments. The incorporation of perspective transformation ensures geometric consistency, while the dual-branch architecture enhances feature extraction and improves segmentation accuracy. The auxiliary prediction mechanism further strengthens the learning process by providing intermediate supervision, enabling the model to capture fine-grained details and improve boundary detection.



REFERENCES

Base Paper (Hyperlink: [IEEE Xplore - SDLMA-IITS](#))

Citation:

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Review Comments	Action taken
Align the equal no of objectives and outcomes	Equal no of objectives and outcomes are placed
Dataset not included in system diagram	Dataset Added
Not cleaned with existing and proposed model	Done
Include review comments and actions taken in presentation slide	Added
Fine tune the objectives and outcomes	Modified successfully
Represent the result in a table format	Result Table Added Successfully



Review Comments	Action taken
Include review comments and actions taken in presentation slide	Added
Fine tune the objectives and outcomes	Modified successfully
Represent the result in a table format	Result Table Added Successfully



THANK YOU